Phinma University of Iloilo
COLLEGE OF INFORMATION TECHNOLOGY EDUCATION
Rizal, Iloilo City, Philippines

GONZAGA DENTAL CLINIC MANAGEMENT AUTOMATION SYSTEM (GDCMAS)

A Business Process Automation Plan for Streamlining Patient Records, Appointment Scheduling, and Dental Clinic Operations at Gonzaga Dental Clinic

A BAM - 285 Final Course Work
Presented to the Faculty of the
College of Information Technology Education
Phinma University of Iloilo
Rizal, Iloilo City

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology

by

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October 2026

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Approval Sheet

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Acknowledgment

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EXECUTIVE SUMMARY

[CONTEXT]

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Chapter I

Introduction

1.1 Background of the Project

Dental clinics manage a range of administrative and clinical activities, including patient registration, appointment scheduling, consent documentation, treatment recording, and retrieval of patient history. These activities require accurate and accessible information so that clinic personnel can coordinate patient visits and support continuity of care. When records are maintained entirely on paper, routine tasks may depend on the physical availability, legibility, and proper filing of documents.

Gonzaga Dental Clinic operates two branches. Based on the initial interview and site observations shared for this study, the Main Branch has two employees, while the Second Branch has one employee. The clinic currently uses a fully manual, paper-based process for patient information and treatment documentation. During patient intake, the secretary provides paper forms for the patient to complete. The secretary explains the terms and conditions, after which the patient signs a physical consent form. The completed forms are placed in a brown envelope labeled with the patient's name.

During consultations, the dentist records treatment information on index cards. Patients sometimes take photographs of these cards to keep a personal copy of their treatment records. The envelopes and index cards are stored in binders or cabinets. When patient information is needed at the other branch,

staff may need to travel physically or share photographs and messages through online chat. This approach makes information exchange dependent on manual coordination and may expose patient information to inappropriate access or unintended disclosure.

The initial interview identified several operational concerns: searching for a patient record requires staff to browse physical files; the two branches do not have instant access to a shared patient record; there is no reported backup copy of the records; duplicate records may be created when existing files cannot be located; and appointments are likely recorded in a logbook, which may be prone to scheduling conflicts or missed entries. These concerns can consume staff time, delay access to patient history, and make consistent record management more difficult.

To address these concerns, this project proposes GDCMAS, a digital dental clinic management and record system for Gonzaga Dental Clinic. The proposed system is intended to centralize patient records across both branches and support digital patient registration, searchable records, treatment history, appointment management, and secure information access. The project will analyze the current process and define a proposed workflow and system requirements. The proposed functions and implementation details will require confirmation with the clinic before development.

1.2 Statement of the Problem

1.2.1 General Problem

How can Gonzaga Dental Clinic improve its manual, paper-based patient information and record management process through a centralized digital system that supports efficient record retrieval, coordinated access across its two branches, secure handling of patient information, and organized appointment management?

1.2.2 Specific Problems

1. Patient records are stored in physical envelopes, binders, and cabinets, requiring staff to search manually when a record is needed.
2. Patient information is not instantly accessible between the Main Branch and the Second Branch, resulting in physical travel or the sharing of record photographs and messages.
3. The clinic has no reported backup system for its physical records, leaving the information vulnerable to loss or damage.
4. Duplicate patient records may be created when staff cannot locate an existing file, potentially fragmenting a patient’s treatment history.
5. Appointment information is likely maintained in a logbook, creating a risk of double-booking, missed appointments, or difficulty checking schedule availability.

6. Treatment updates are recorded on index cards, and patients may rely on photographs of these cards to retain a personal copy, rather than accessing a consolidated digital history.

1.3 Objectives of the Project

1.3.1 General Objective

To analyze and design a proposed Business Process Automation (BPA) plan and centralized digital dental clinic management and record system for Gonzaga Dental Clinic that improves patient record retrieval, supports coordinated operations across two branches, strengthens information management, and organizes appointment scheduling.

1.3.2 Specific Objectives

1. Document the clinic's existing patient intake, consent, record creation, treatment documentation, storage, and inter-branch information-sharing processes.
2. Identify the operational problems, process gaps, and information-management risks associated with the current paper-based workflow.
3. Identify the system stakeholders and determine their responsibilities and initial system needs.
4. Model the current process using an As-Is workflow and analyze the process using SIPOC and root-cause analysis.
5. Define the functional and non-functional requirements for a centralized digital system, subject to validation by the clinic.

6. Design a proposed To-Be workflow for digital patient registration, consent documentation, record retrieval, treatment updates, and appointment management.
7. Develop a proposed system structure and modules that address the identified operational needs.
8. Prepare a preliminary testing and implementation plan for future validation and deployment.

1.4 Significance of the Project

Gonzaga Dental Clinic: The proposed system may provide a more organized way to manage patient records and appointments across both branches. Centralized records can reduce dependence on physical file transfer and support consistent information access.

Clinic Owner / Dentist: The system is intended to make patient treatment history easier to retrieve and update, supporting review of previous dental procedures during patient visits. Access controls would help limit record access to authorized users.

Secretary and Clinic Staff: Digital registration, searchable patient profiles, and an organized appointment calendar may reduce time spent handling paper forms, locate envelopes, and check logbooks.

Patients: Patients may benefit from more organized registration, reduced repetition of information, and improved coordination when receiving services at either branch. Any patient-facing access or digital consent feature will depend on the clinic's final requirements.

Future Researchers and Developers: The project may serve as a reference for analyzing and designing a small, multi-branch dental clinic management system using business process analysis methods.

1.5 Purpose and Description

The purpose of GDCMAS is to provide a proposed digital alternative to the clinic's current paper-based workflow. The system is envisioned as a centralized application that allows authorized clinic personnel to create and search patient profiles, maintain treatment records, manage appointment schedules, and access patient information across the two branches. Instead of relying exclusively on brown envelopes, index cards, and logbooks, the clinic would maintain structured digital records in a shared database.

The proposed workflow begins with patient registration or retrieval of an existing profile. The clinic staff records the required patient details and consent status. During or after a consultation, the dentist or authorized staff member updates the patient's treatment record. Appointment information is recorded in a shared calendar, allowing staff to check existing schedules before assigning a slot. When a patient visits either branch, authorized personnel can search for the patient's existing record, subject to the system's access rules.

The system is a proposal based on the initial interview information. The final design should be confirmed through additional consultation with the clinic, particularly regarding required patient fields, consent wording,

treatment-chart format, appointment rules, user roles, retention practices, and backup arrangements.

1.6 Scope and Limitations

Scope

1. Patient profile registration and maintenance, including a searchable patient directory.
2. Centralized access to patient records for the Main Branch and the Second Branch, with access restricted to authorized users.
3. Digital storage and retrieval of patient intake information and consent documentation, subject to clinic approval of the forms and process.
4. Recording and updating of dental treatment history, including the information currently documented on index cards.
5. Appointment scheduling and calendar viewing to help staff check available and occupied time slots.
6. Basic user and role management for clinic personnel.
7. Backup planning and basic audit logging to support record availability and accountability.

Limitations

1. The proposed system does not replace the dentist’s professional judgment, diagnosis, or treatment decisions.

2. Automated diagnosis, treatment recommendations, and AI-based interpretation of dental images are not included in the initial scope.
3. Integration with dental imaging equipment, laboratory systems, insurance providers, payment gateways, or government health systems is not established and will require separate requirements and feasibility assessment.
4. The system's final hosting platform, budget, development technology, backup frequency, and recovery targets have not yet been confirmed with the client.
5. Existing paper records may require manual encoding or scanning. The completeness and accuracy of any migrated records will depend on the condition of the source documents and validation by clinic personnel.
6. Actual performance improvements, user acceptance, and security effectiveness cannot be claimed until the system is developed, tested, and evaluated.

Chapter II

REVIEW RELATED LITERATURE AND SYSTEMS

2.1 Related Literature

Sidek and Martins [1] examined the factors associated with electronic health record (EHR) implementation in a dental clinic context. Using Grounded Theory and data from the implementation context of Brunei's national Healthcare Information and Management System, the study identified six perceived critical success factors: system usability, emergent behaviours, requirements analysis, training, change management, and project organisation. The authors also reported that workflow changes could challenge clinicians' confident use of an EHR, particularly when systems have limited modularity and configurability. These findings are relevant to GDCMAS because the proposed system changes the clinic's existing paper-based workflow. The study supports the need to conduct requirements analysis with actual users, consider usability, and provide training and change-management activities before full adoption [1].

Suebnukarn et al. [2] developed and assessed the usability of an electronic health record system in a comprehensive dental clinic. The system interface was designed using cognitive ergonomics, and cognitive task analysis was used to examine the tasks performed by EHR users. The study evaluated 30 oral-examination cases involving ten EHR users and found that users needed an average of 27 steps to complete the evaluated tasks for one case. The researchers concluded that the user interface could be improved by reducing the mental effort required to complete tasks. This study is relevant

to GDCMAS because the proposed system should not only digitize records but also provide a simple and efficient interface for secretaries and dental personnel [2].

Farajollahi and Sheikhtaheri [3] evaluated the content and features of electronic dental record (EDR) systems in 22 dental clinics in Tehran, Iran. The researchers compared EDR data elements with standards and evaluated system features using established health-information-system models. Their study identified 219 data elements and 74 system features, while also finding that some clinical data elements and advanced features were absent across the systems examined. The authors concluded that existing EDR systems may function primarily as administrative and information-recording tools and recommended greater attention to standardized clinical data and decision-support capabilities. For GDCMAS, this study highlights the importance of defining the required patient, treatment, appointment, and administrative information before system development so that the proposed database reflects the actual needs of Gonzaga Dental Clinic [3].

Ramadhan [4] designed and implemented a web-based electronic medical record system for Bertha Desnalia Dental Clinic. The clinic previously managed patient registration, medical records, queues, drug inventory, and billing manually. The study identified problems including data redundancy, slow retrieval of patient history, and error-prone manual billing. The system was developed using the Waterfall model with PHP, CodeIgniter, and MySQL and included authentication, patient registration, queue management, medical records, drug management, and billing. Functional black-box testing was

conducted on six modules, and the predefined test scenarios produced the expected outputs. The authors were careful to note that reductions in retrieval time and billing errors were design goals rather than measured outcomes because comparative timing and error-rate measurements were not collected. This study is closely related to GDCMAS because both projects address the transition from manual dental-clinic processes to a web-based digital record system [4].

The World Health Organization describes an electronic medical record as a secure digital system containing health and clinical-care information managed by healthcare providers. Digital records can support structured patient information, clinical documentation, and other healthcare information-management activities. This provides a broader digital-health foundation for GDCMAS, where patient registration, treatment history, and appointment information are proposed to be maintained in a centralized digital environment [5].

Patient information requires appropriate privacy and security controls. In the Philippines, Republic Act No. 10173, or the Data Privacy Act of 2012, provides the legal framework for the protection of personal information. For GDCMAS, this supports the inclusion of authentication, role-based access, controlled handling of patient records, and appropriate safeguards for information shared between the clinic's two branches. The final privacy, retention, consent, and access procedures should be validated against the clinic's actual operations and applicable requirements [6].

2.2 Related Systems

Open Dental is a dental practice management platform that provides functions including digital forms, a graphical tooth chart, patient portal, web scheduling, secure email, and automated messaging. These features demonstrate how dental-clinic administrative and clinical activities can be brought together in a digital platform. GDCMAS similarly proposes patient records, treatment documentation, and appointment management, but its scope is specifically based on the operational requirements identified at Gonzaga Dental Clinic [7].

Dentrix Enterprise is designed for dental practices operating across multiple locations. Its published information describes centralized patient information, electronic records, scheduling, and cross-site access. These capabilities are relevant to GDCMAS because Gonzaga Dental Clinic operates two branches and currently relies on physical transfer or online sharing when records are needed at another location. GDCMAS is proposed specifically for the clinic's identified needs and does not claim feature or cost equivalence with Dentrix Enterprise [8].

Dentrix PH presents a cloud-based dental clinic management platform with functions related to patient records, dental charting, scheduling, billing, and reporting. Its feature set provides another example of integrating several dental-clinic activities into a digital system. GDCMAS uses a similar integrated approach while focusing on the paper-based processes and multi-branch record-access requirements identified during the Gonzaga Dental Clinic assessment [9].

2.3 Synthesis

The reviewed literature and systems demonstrate the increasing use of electronic health and dental record systems to organize patient information, improve record accessibility, and support the management of dental clinic operations. Sidek and Martins [1] emphasized the importance of requirements analysis, usability, training, change management, and organizational support when implementing electronic health records in a dental clinic. Similarly, Suebnukarn et al. [2] examined the usability of an electronic health record in a dental clinic and highlighted the importance of designing the system around the tasks and needs of its users.

Farajollahi and Sheikhtaheri [3] examined the content and features of electronic dental record systems and identified the importance of defining appropriate patient and clinical data elements and system features. This supports the need for GDCMAS to establish clear requirements for patient profiles, treatment history, appointment information, and other clinic records before development. Ramadhan [4] further demonstrated the applicability of a web-based electronic medical record system in a dental clinic that previously relied on manual processes. The study included functions for patient registration, medical records, queue management, and billing, which are relevant to the digital transformation proposed for Gonzaga Dental Clinic.

The reviewed systems also demonstrate that dental practice management platforms can integrate several clinic activities into a centralized environment. Open Dental [7] provides functions such as digital forms,

patient records, scheduling, and dental charting, while Dentrix Enterprise [8] supports centralized information management and access across multiple locations. Dentrix PH [9] similarly demonstrates the integration of patient records, dental charting, scheduling, billing, and reporting in a digital dental clinic environment. These systems provide references for identifying possible functions that may be considered in the design of GDCMAS.

However, the reviewed studies and existing systems are not designed specifically around the current processes and requirements of Gonzaga Dental Clinic. The clinic operates two branches and currently relies on paper forms, brown envelopes, index cards, physical filing, and manual information sharing between branches. Therefore, GDCMAS is proposed to address these specific process gaps through centralized patient records, digital treatment documentation, appointment management, controlled branch access, and organized record retrieval.

Overall, the reviewed literature and systems provide a foundation for the development of GDCMAS. The literature emphasizes usability, requirements analysis, user involvement, electronic record management, and proper implementation, while existing dental management systems demonstrate how patient records, treatment documentation, scheduling, billing, and reporting can be integrated into a digital platform. These findings support the proposed direction of GDCMAS while allowing its final features and workflows to remain specific to the validated needs of Gonzaga Dental Clinic.

Chapter III

METHODOLOGY, RESULTS, AND DISCUSSION

3.1 Related Systems

The requirements analysis is based on the initial interview information and site observations provided for Gonzaga Dental Clinic. The analysis documents the current workflow, identifies reported operational concerns, and translates those concerns into preliminary system requirements. Since the available information represents an initial assessment, the requirements and proposed workflow should be validated with the clinic before they are treated as final.

3.1.1 Stakeholder Analysis

The initial stakeholders are identified according to their known involvement in the clinic's operations. The owner/dentist and secretary are directly supported by the interview description. The patients are included because they provide registration information and consent and receive dental services. A separate administrator role is proposed only if the clinic assigns someone to manage user accounts.

Table 1. Stakeholder Analysis Matrix

Stakeholder	**Role / Responsibilities**	**Current Needs or Pain Points**	**Proposed System Expectations**
Clinic Owner / Dentist (Dr.	Oversees clinic operations and	Treatment information is	Review and update patient

Lezly S. Gonzaga)	provides dental care; updates treatment records.	written on index cards; records may need to be shared between branches.	treatment history; access records across branches; maintain appropriate control over clinical information.
Secretary / Clinic Staff	Provides intake forms, explains terms and conditions, collects signatures, files records, and likely manages appointments.	Manual form handling, physical filing and searching, and coordination when records are needed at the other branch.	Register and search patients; record consent; manage appointments; retrieve authorized patient records efficiently.
Patients	Provide personal information,	May need to provide	A more organized

	complete forms, sign consent, and receive dental services.	information on paper; some take photographs of index cards to retain treatment information.	registration and record process; clear consent handling and appropriate protection of personal information.
System Administrator (proposed, if assigned)	Manages user accounts and system configuration.	No current digital administration process was described.	Create or deactivate accounts, assign access permissions, and support backup and basic system maintenance.

3.1.2 SIPOC Analysis

The Supplier-Input-Process-Output-Customer (SIPOC) model summarizes the high-level patient record management process. The current-state column reflects the interview details; the proposed outputs describe the intended digital workflow.

Table 2. SIPOC Analysis Framework

Supplier	Input	Process	Output	Customer
Patient	Personal and contact details; completed intake form; signed consent	Provide information and complete registration and consent requirements	Patient information and consent record	Secretary / Dentist
Secretary	Paper forms; patient-provided information	Provide forms, explain terms and conditions, collect signatures, and create a labeled envelope	Completed intake documents and patient file	Dentist / Clinic
Dentist	Patient file; index card; consultation information	Review patient history and manually update treatment	Updated treatment index card	Patient / Clinic

		details on an index card		
Clinic staff at either branch	Patient name; paper record or photos/messages shared through chat	Search physical files or coordinate record transfer / sharing	Retrieved or shared patient information	Dentist / Secretary
Clinic owner / authorized staff (proposed)	Patient database; appointment details; user permissions	Search records, update authorized information, and manage appointments in the proposed system	Centralized patient record and appointment schedule	Both branches / Patients

3.1.3 Related Systems

Based on the initial interview, the clinic's current process begins when the secretary provides physical forms to a patient. The patient completes the forms manually. The secretary explains the terms and conditions, and the patient signs a paper consent form. The completed documents are placed in a brown envelope labeled with the patient's name. During treatment, the dentist records updates on index cards. The envelope and index card are then stored in binders or cabinets. When the record is needed at the other branch, staff may physically travel or send photographs and messages through online chat.

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CHAPTER IV

RESULTS AND DISCUSSION

4.1 Implementation

This chapter presents the results of the development and implementation of the Gonzaga Dental Clinic Management Automation System (GDCMAS). The system was developed using Next.js 16, React 19, TypeScript, Tailwind CSS 4, and Supabase as the backend database and authentication provider.

The implementation focused on digitizing and automating the clinic's core business processes, including patient registration, appointment scheduling, treatment recording, consent management, and inter-branch operations.

4.2 Development Tools and Technologies

The following tools and technologies were utilized in the development of GDCMAS:

Category	Tools/Technologies
Frontend Framework	Next.js 16, React 19, TypeScript
Styling	Tailwind CSS 4
Backend/Database	Supabase (PostgreSQL, Auth, RLS)
Validation	Zod
Development Environment	Visual Studio Code
Version Control	Git and GitHub

4.3 Hardware and Software Requirements

4.3.1 Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i5 or equivalent
RAM	8 GB or higher
Storage	256 GB or higher
Internet Connection	Stable broadband connection

4.3.2 Software Requirements

Software	Requirement
Operating System	Windows 10/11, Linux, or macOS
Web Browser	Google Chrome, Mozilla Firefox, or Microsoft Edge
Node.js	Version 18.18.0 or higher
Database	PostgreSQL 15+ (via Supabase)

4.4 System Features and Modules

• Patient Management Module

• Appointment Scheduling Module

• Treatment Records Module with FDI Tooth Chart

• Consent Management Module

• Public Booking Portal

• Staff Management Module

• Branch Management

• Audit Logging Module

4.5 Security Implementation

GDCMAS implements multi-layered security: Row Level Security on all 9 tables, three-layer authorization (Middleware $\rightarrow$ Server $\rightarrow$ RLS), role-based access control, branch-level isolation, audit logging, and database constraints preventing invalid operations.

4.6 Results Analysis

The system successfully addresses all identified problems. Centralized records eliminate fragmentation, automated scheduling prevents conflicts, role-based access ensures security, and audit logging provides accountability across both branches.

4.7 System Evaluation

The implementation meets all proposed objectives. Core modules are functional, security measures are enforced at the database level, and business processes are successfully digitized.

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CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary, findings, conclusions, and recommendations based on the development and implementation of GDCMAS.

5.2 Summary of the Study

The study aimed to develop and implement GDCMAS to automate Gonzaga Dental Clinic's business processes. Problems in the paper-based system were identified and addressed through a secure, web-based solution using modern technologies.

5.3 Findings

1. Paper-based system caused fragmentation and inefficiency.

2. Manual booking led to conflicts and duplicate records.

3. No backup or audit trail existed.

4. GDCMAS successfully centralized records with role-based access.

5. Multi-layered security and RLS protect sensitive data.

6. Database constraints ensure data integrity.

7. Audit logging ensures accountability.

5.4 Conclusion

GDCMAS effectively addresses all identified problems. It improves efficiency, accessibility, security, accountability, and patient care quality. The system provides enterprise-grade security while enabling seamless inter-branch operations, establishing a solid foundation for digital transformation.

5.5 Recommendations

1. Provide staff training on system usage.

2. Maintain regular backups via Supabase.

3. Deploy with proper domain and SSL.

4. Conduct periodic security reviews.

5. Consider future enhancements (notifications, reports, inventory).

6. Establish clear user management policies.

7. Perform regular updates and maintenance.